

Multiple Choice

1. **Answer:** A

Options: A) $f(n) = [3 (1/2)^n + 1 (1/3)^n + 1] u[n]$; B) $f(n) = [4 (1/2)^n - 1 (1/3)^n] u[n]$; C) $f(n) = [3 (1/2)^n - 2 (1/3)^n] u[n-1]$; D) $f(n) = [3 (1/2)^n + 2 (1/3)^n] n[n]$; E) $f(n) = [2 (1/2)^n + 3 (1/3)^n] u[n]$

Note: Correct option: A - $f(n) = [3 (1/2)^n + 1 (1/3)^n + 1] u[n]$

2. **Answer:** A

Options: A) Number of isotopes.; B) Atomic number.; C) The element's period on the periodic table.; D) Column number on periodic table.; E) Atomic radius.

Note: Correct option: A - Number of isotopes.

3. **Answer:** E

Options: A) 100 ohms; B) 25 ohms; C) j 100 ohms; D) 50 ohms; E) - j 50 ohms

Note: Correct option: E - - j 50 ohms

4. **Answer:** A

Options: A) $0.075 \lambda^2$; B) $0.200\lambda^2$; C) $0.180\lambda^2$; D) $0.089\lambda^2$; E) $0.102\lambda^2$

Note: Correct option: A - $0.075\lambda^2$

4. **Answer:** D

Options: A) $f(x) \sim (1/2)c + (2c/\pi^2)\sum_k = 1[\{\cos[(2k)/c]\}/(2k)^2]$; B) $f(x) \sim (1/2)c + (4c/\pi^2)\sum_k = 0[\{\cos[(2k+1)/c]\}/(2k+1)^2]$; C) $f(x) \sim (1/2)c - (4c/\pi^2)\sum_k = 1[1/n^2]\cos\{(n)/c\}$; D) $f(x) \sim (1/2)c - (4c/\pi^2)\sum_k = 0[\{\cos[(2k+1)/c]\}/(2k+1)^2]$; E) $f(x) \sim (1/2)c - (4c/\pi^2)\sum_k = 1[\{\cos[(2k)/c]\}/(2k)^2]$

Note: Correct option: D - $f(x) \sim (1/2)c - (4c/\pi^2)\sum_k = 0[\{\cos[(2k+1)/c]\}/(2k+1)^2]$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '2.25 per cent'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'a' = $2 RT_c V_c$, b' = $(V_c/2)$ '.

9. **Answer:** False

Options: A) True; B) False

Note: Actual answer is '100 ohms'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '5,000 pF'.

Long Form Response

11. **Answer:** 166.7 V and 133.3 V. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 26.1 %. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key